

Field Service Memo

MR

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Announcement about the tool software log for decoding Superconducting magnet information

1. Abstract

This document announces the release of the Service tool software “Magnet Log Viewer V3.15.15.9^(*A)” of log decoding for the superconducting magnet, and explains how to use it.

2. Applicable system

All system equipments OR97, OR200, TN150 Revision none (1st – 5th lot), TN150 Revision A (6th lot), TN150 Revision B (7th lot) ^(*A)

3. Applicable date

The same day of this FSM issuing

4. Contents

To analyze the superconducting magnet log (below, Mag log) which is generated from SVU or MSUP and stored to SP, it is required to convert each parameter and check how to read the log, so spend many time trouble shooting. For the sake of reduction of analyzing load and time, we have developed the tool software “Magnet Log Viewer”, which can decode the log generated from SVU or MSUP, and displays its condition visually.

This FSM announces its release and explains how to use it below.

4.1. Purpose of this tool software

- ✓ It can decode Mag log, generated to SP, or downloaded to service PC from SP.
- ✓ It can assist CE's understanding of the target Cooling Cabinet's condition, by displaying the decode result visually easy-to-understand.
- ✓ This tool software is just a viewer for assistance of CE's understanding. Therefore, below functionality's are not applicable.
 - Automatic judgement about ordering applicable service parts.
 - Automatic/Manual downloading functionality of Mag log from itself or SP.
 - Other functionalities, deviating from the above purpose.

4.2. Execution environment

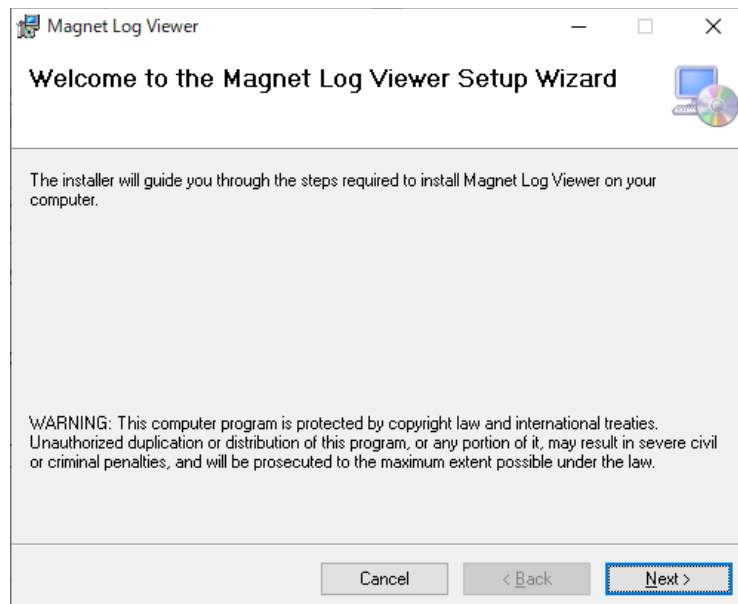
Please note to execute Magnet Log Viewer under execution environment below. Magnet Log Viewer has been confirmed to be able to execute and perform in service PC and SP with listed Windows and software below.

- ✓ Windows10 64bit (Japanese or English) OR
Windows7 64bit / 32bit (Japanese or English)
- ✓ **The latest version's Log Analyzer, which has installed the valid license.** (*A)
- ✓ An environment where all the latest Windows Update have been applied.

4.3. Installation

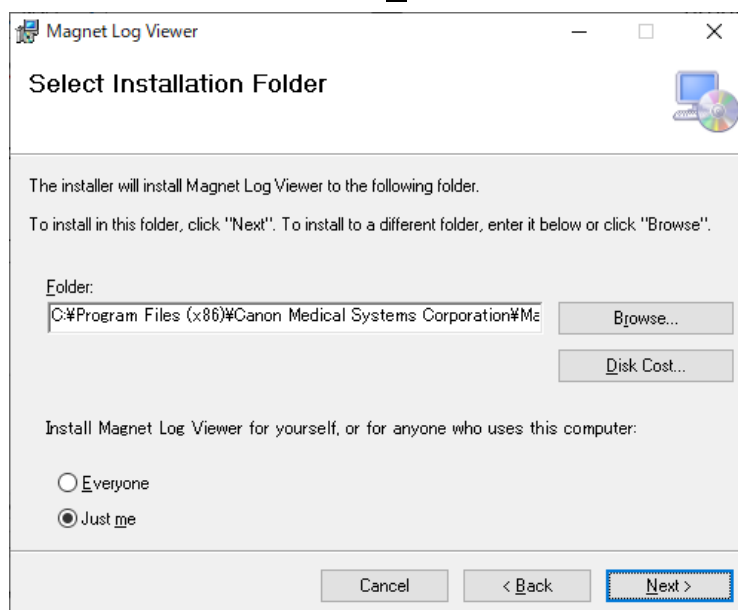
After download the Installer of Magnet Log Viewer (below, MagLogViewer.msi) from FIXS knowledge, please install following below steps.

- (1) Double-click or select CoolCabiLogViewer.msi and press [Enter] key.
After showed below dialog, click [Next]^(*A) on right-below of dialog.



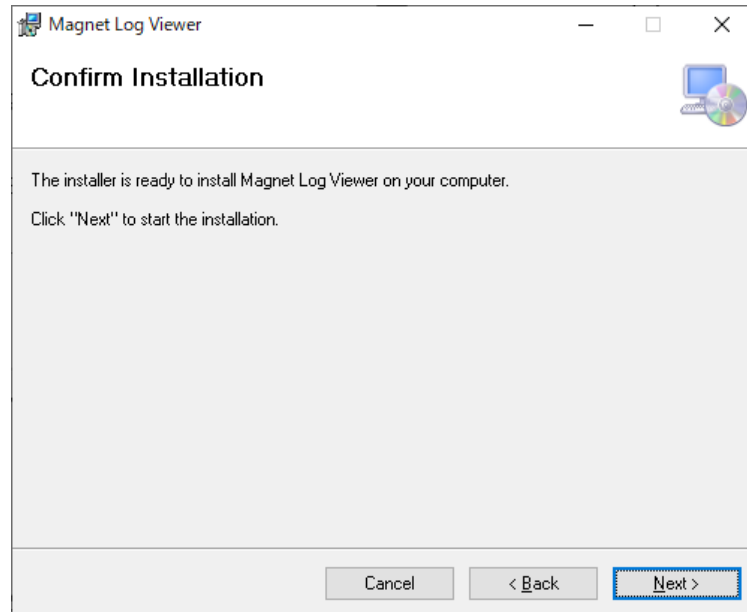
Installation entry screen of Magnet Log Viewer

- (2) On "Select install folder" and selection applicable user below, **after confirmation that the same directory with installation directory of LogAnalyzer has been selected**, click [Next]^(*A) on right-below of dialog.



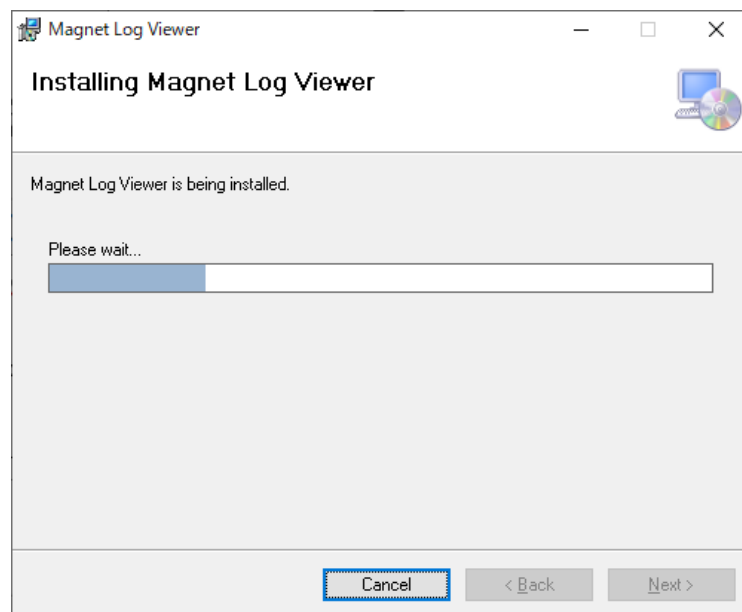
The screen of "Select install folder" and of user selection

- (3) On below screen, select [Next]^(*A) on right-below of dialog, start installation.



Confirmation screen before starting installation

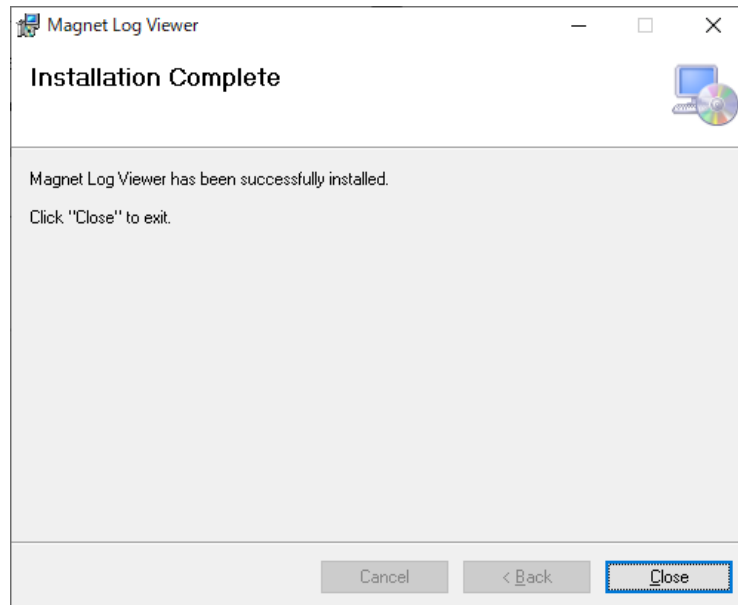
- (4) If installer starts installation, screen becomes like the example below, and progress bar under "Please wait..." message proceeds. Progress bar may be updated several times, so please wait for a while.



Installation in progress screen

- (5) If below screen is displayed, installation has been done successfully, so click [Close]^(*A) on right-below of dialog, and finish installation.

If below screen is not displayed, installation may have been failed, so restart installation steps from start (Section 4.2). If it doesn't become the screen shown below after restart installation, report phenomenon and detail steps can be found referring "5. Notes".



Installation success screen

- (6) If installation has been done successfully, short-cut icon will be generated on Desktop, invoke Magnet Log Viewer from here.

Execution directly binary file pointed from short-cut is not recommended, because it may be affected on any other tool software which are placed on the same directory.



Short-cut on desktop

4.4. Invocation

Magnet Log Viewer can be invoked two ways.

If the valid log file is read, discerning TN150, OR97, or OR200 automatically, tool software will be invoked. If tool software cannot invoke, an error description pop-up dialog may be displayed.

✓ **By selecting Mag log file**

If you double-click the short-cut of Magnet Log Viewer, "Open file" dialog will be displayed. Select expected log file from the dialog, click [Open] on right-below of dialog.

Either ".txt" or ".csv" extensions are selectable from the drop-down list beside of [File Name] in this dialog. ".txt" is default. Please select match extension from this drop-down list, if extension of analyzed file was ".csv". (*A)

For your information, this "Open file" dialog doesn't support Drag & Drop, although the short-cut icon of this tool software supports it.

✓ **By Drag & Drop Mag log file**

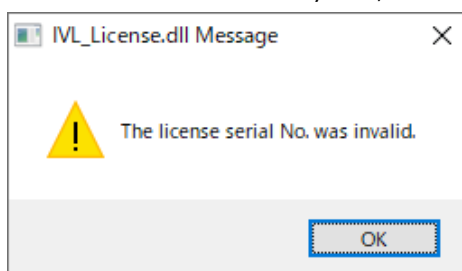
Drag & Drop the expected Cooling Cabinet log file on its short-cut.

CAUTION

- (1) Log file name opening by Magnet Log Viewer should be "XXXX_SVUlog.txt (for TN150 Revision none)", "XXXX_MSUP_VC.txt (for OR97)", "XXXX_MSUP2.txt (for OR200)", "XXXXsvu_full_logXXXX.csv (or .txt)", "XXXXsvu_long_logXXXX.csv (or .txt)", or "XXXXsvu_short_logXXXX.csv (or .txt)". (*A) Magnet Log Viewer cannot invoke without following this naming rule.
- (2) If any log format which Magnet Log Viewer cannot understand is/are described in log file, the tool software cannot invoke.
- (3) If LogAnalyzer license is invalid (old, corrupted, copied, no contents, etc...), or tool software cannot evaluate LogAnalyzer license in correct (incorrect license directory, invalid license evaluation, etc.), the tool software cannot invoke.

NOTE

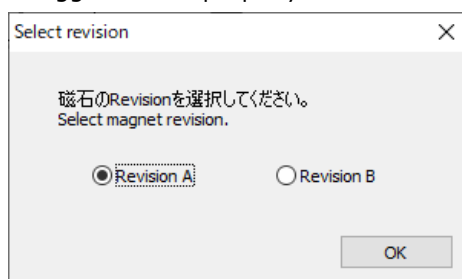
In case of invalidated (ex. expired) LogAnalyzer license (below, license) existence, pop-up dialog like below will be displayed. This pop-up dialog is not shown, if invalidated licenses are deleted. And tool software can invoke, if validate license exists even only one, even if this pop-up dialog is shown.



Pop-up dialog which will be displayed in case of being installed invalidated license

NOTE^(*A)

The below dialog will appear, in case Mag log file generated from TN150 Revision A or TN150 Revision B is opened. Please select the button suitable correct magnet revision. If it does not match, the software is unable to evaluate logged values properly.

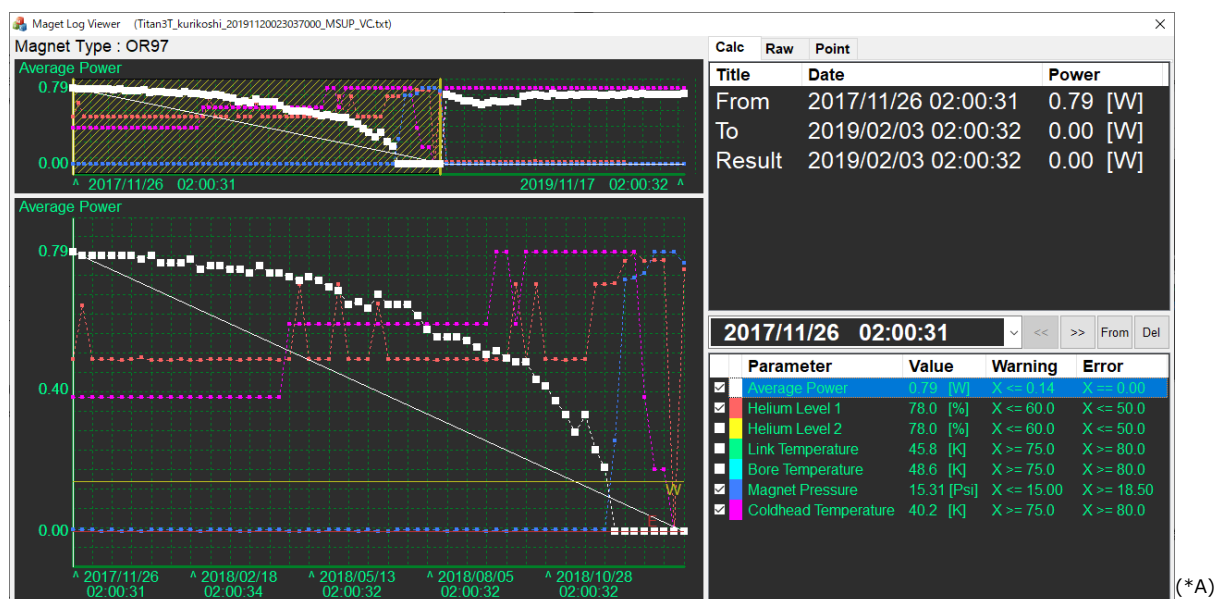


Pop-up dialog which will be displayed in case of opening log file generated from TN150 Revision A or TN150 Revision B

4.5. Running screen and operation

If Magnet Log Viewer can be invoked successfully, the screen like bellows example will be displayed.

Here displays an example of reading Mag log file generated from OR97^(*A).



A case example of reading and invoking the log file generated from OR97^(*A)

The displayed list, graph and tab contents are changed depend on the superconducting magnet type, according to below table ^(*A).

Supported parameters for each magnet

Magnet	TN150 Revision none	TN150 Revision A/B (Full)	TN150 Revision A/B (Long)	TN150 Revision A/B (Short)	OR97	OR200
Parameter						
Helium Level	✓	✓	✓	✓		✓
Helium Level 1					✓	
Helium Level 2					✓	
Shield Temperature	✓	✓	✓	✓		
Link Temperature					✓	✓
Bore Temperature					✓	✓
Magnet Pressure	✓	✓	✓	✓	✓	✓
Cold Head Temperature		✓	✓			✓
Average Power					✓	✓
Heater Power		✓	✓			

These are not supported: OR76、TN150 Revision A or B (RMU)、TN150 Revision B (ERDU)

The meaning of each part and the operation are as below.

✓ Forward (>>) button

It can proceed log time of displaying. If the end row of log is selected, this button become disable.

This functionality is available by left-clicking or wheeling at any place where mouse cursor changes to cross on each graph area. (*A)



Forward (>>) button

✓ Rewind (<<) button

It can rewind log time of displaying. If the first row of log is selected, this button become disable.

This functionality is available by left-clicking or wheeling at any place where mouse cursor changes to cross on each graph area. (*A)

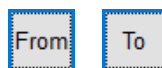


Rewind (<<) button

✓ Range setting button

On Main graph and Sub graph (these are explained later), it can select the ROI of the red log. Displayed "From" means you can set ROI start, and displayed "To" means you can set ROI end. These caption changes in each clicking. You can clip the ROI of log, by moving expecting date on "Date and time list", and clicking this button on each date.

This functionality is available by right-clicking at any place where mouse cursor changes to cross on each graph area. (*A)



Range setting button (Left: ready for start setting, right: ready for end setting)

✓ Range deselection button

It deselects selected ROI of log. This become disable in case ROI of log hasn't selected once yet.



Range deselection button

✓ Date and time list

It displays current selected log date and time. Clicking “v” mark on the right displays pull-down list, and enable to select any date. And mouse wheel, up key and down key, Fwd (>>) button and Rwd (<<) button, can feed date forward/backward, also.

This functionality is available by left-clicking or wheeling at any place where mouse cursor changes to cross on each graph area. (*A)



Date and time list

✓ Parameter list

It displays that, each parameter name which shows superconducting magnet status, measured value, and thresholds in string, evaluated result according to each threshold in color. From left, check box for graph display/hide, color legend, parameter name, measured value, Warning threshold, and Error/Alert threshold, are displayed respectively.

Color means Normal as green, Warning as yellow, and Error / Alert as red. Gray color means non-numeric and invalid log (ex. “----”). Please take case that non-numeric and invalid log will be treated and displayed as 0 in each graph for the sake of simplicity. (*A)

If you click any column, it becomes selected condition. If you click blank area in the lowest part of list, it becomes non-selected condition. The vertical axis on Main graph and Sub graph (these are explained later) shows values about selected parameter. And waveform plots of selected parameter are displayed larger than another parameter’s waveform.

Check mark can switch by clicking each check boxes. According to On/Off of check mark, graph display/hide is also switched. Please note check mark at selected parameter cannot be off. And check mark put on automatically if unchecked parameter is selected.

Parameter which are check marked at invocation of tool software, and displaying parameter type, are changed depending on the read log.

	Parameter	Value	Warning	Error
☒	Helium Level	5.0 [%]	X <= 52.0	X <= 45.0
☒	Shield Temperature	--- [K]	X >= 60	X >= 65
☒	Coldhead Temperature	5.3 [K]	X >= 84.0	X >= 102.0
☒	Magnet Pressure	8847 [Pa]	X <= 400	X >= 3000
☒	Heater Power	0.03 [W]	X <= 0.10	X == 0.00

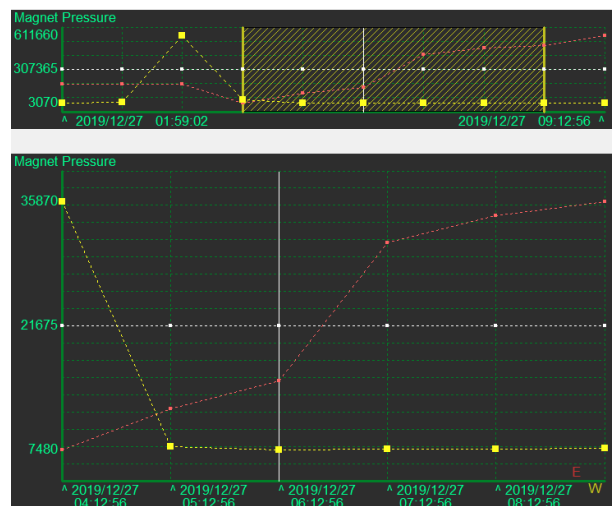
(*A)

Parameter list (A case example analyzing TN150 Revision A^(*A) log)

✓ Graph

■ Main graph / Sub graph

It displays graph plotting all measured value about parameters marked on Parameter list within all logged magnet parameters in log file. Downside big graph area (Main graph) displays the selected ROI of upside narrow graph area (Sub graph) with expansion.



Graph (Upside: Sub graph, Downside: Main graph)

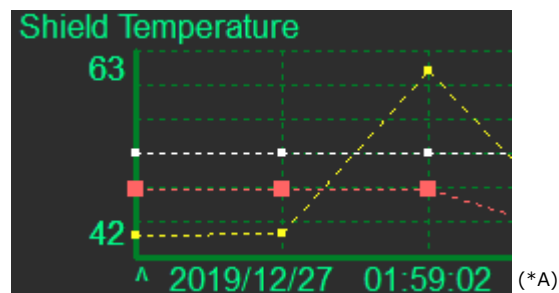
■ Waveform, vertical axis, and horizontal axis in graph

It displays Measured value as dot, and grid line connected each dot. If nothing is selected in Parameter list, only vertical axis and horizontal axis are displayed. Only selected parameter's wave has the larger plot.

Vertical axis shows measured value, and each value means that, Selected parameter name on Parameter list, Maximum of measured value, Intermediate

of measured value, and Minimum of measured value of Graph area, from upside. Vertical axis is scaled automatically like Maximum and Minimum of measured value about the parameter which is selected in Parameter list are included enough in Graph area. Please note parameter name and values on vertical axis displays the contents of selected parameter in Parameter list. And, vertical axis displays ranged Maximum, Intermediate, and Minimum of measured value, if Range setting has been done.

Horizontal axis shows the start and last date (on Sub graph) or 1/6 - 5/6 of all logged date inside of log file.

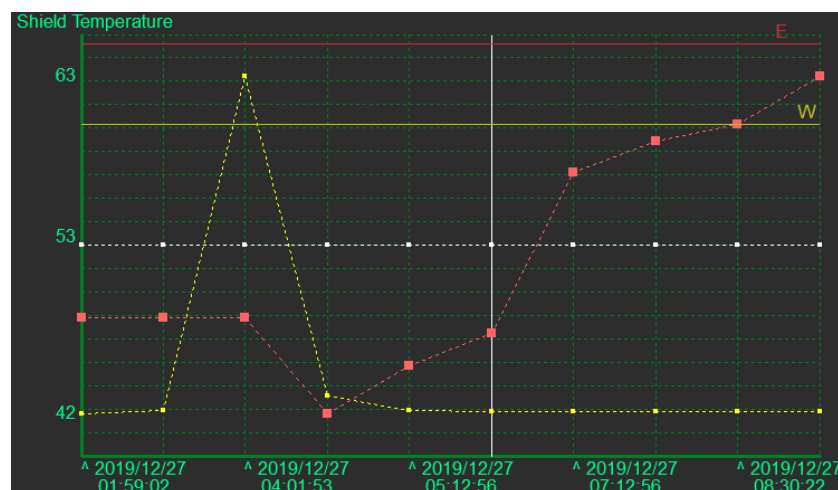


Waveform, vertical axis, horizontal axis (A case example on Sub graph of selecting Shield temperature)

■ Threshold line and current line

Horizontal yellow line with "W" and red line with "E" mean each threshold of the parameter which is selected in Parameter list. If threshold is/are out of graph area, only "W" and/or "E" is/are displayed at upper and/or lower end of graph area. Threshold line is/are displayed only on Main graph.

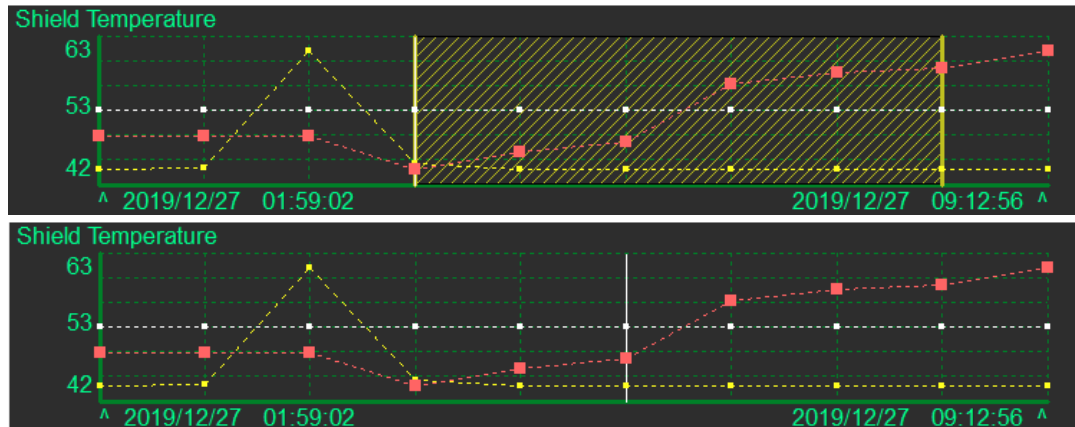
Current Line shows that, position inside of graph area corresponding with line number of current displayed log. It moves in changing displayed date on Date and time list.



Threshold line and current line (An example on Main graph)

■ ROI range setting

In case ROI range setting has done, hatching as range setting are and vertical line as terminal of range, are displayed. This is not displayed right after invocation this tool software and clicking Range deselection button, because ROI range setting hasn't been done.



ROI range setting display on Sub graph (Upside: in case of range set, Downside: right after invocation of tool software or clicking range deselection button)

✓ Status list

It displays corresponding error status, if there is/are Warning or Error/Alert status parameter in the selected date. It displays nothing in case all parameter is within normal range.

Selected lines are copiable by "Ctrl + C" shortcut. (*A)

This list appears only in opening SVU Mag log. (*A)

Status	Raw	Point
No	Status	
1	Pressure Limit High Error	

Status list (A case example of abnormal magnet pressure occurrence)

✓ Raw data list

It displays, all measured value which the invoked tool application reads and analyze from Mag log, as list format. Each parameter string length are longer than cell width, so please widen if required.

Selected lines are copiable by "Ctrl + C" shortcut. (*A)

Status	Raw	Point			
Day	Time	Helium ...	Shield T...	Magnet ...	
2019/12/27	01:59:02	70.4	48	3070	
2019/12/27	04:01:33	70.4	48	8670	
2019/12/27	04:01:53	70.4	48	611680	
2019/12/27	04:12:56	70.4	42	35870	
2019/12/27	05:12:56	70.4	45	7840	
2019/12/27	06:12:56	70.4	47	7480	
2019/12/27	07:12:56	70.4	57	7580	
2019/12/27	08:12:56	70.4	59	7590	
2019/12/27	08:30:22	70.4	60	7610	
2019/12/27	09:12:56	70.4	63	7680	

Raw data list (A case example of reading SVU log)

✓ Check point list

It displays Check point corresponding error status, if there is/are Warning or Error/Alert status parameter in the selected date, or there is/are error status(es) recorded by SVU or MSUP in log file (*A). It displays nothing in case all parameter is within normal range.

Selected lines are copiable by “Ctrl + C” shortcut. (*A)

Status	Raw	Point	
No	Status		
13	[Pressure Limit High Error]		
13-1	Check magnetic field of the mangnet		
13-2	Check compressor static and dynamic pressure		
13-3	Check gas line fittings and oil contamination		
13-4	Check water temperature		
13-5	Check pressure gauge on the magnet		
13-6	Check average power		
13-7	Replace adsorber		
13-8	Replace coldhead		
13-9	Send SPR with MSUP logs to HQ and wait		

Check point list (A case example of abnormal magnet pressure occurrence)

✓ Magnet manufacturing information list (*A)

It displays the characteristic parameters about the magnet from the factory.

Selected lines are copiable by “Ctrl + C” shortcut.

This list appears only in opening SVU Mag log.

Calc	Info	Raw	Point
Helium Parameters			
Probe1: 0% = 0431 = 103.0 Ohms : 100% = 0031 = 7.4 Ohms			
Sample Height = 256 = 250 mA			
Probe2: 0% = 0431 = 103.0 Ohms : 100% = 0031 = 7.4 Ohms			
Sample Height = 256 = 250 mA			
Heater Resistance = 20.5 Ohms			
Z2 Shim Sensitivity = 19.00 ppm/A			

Magnet manufacturing information list

✓ Forecast list of time for cold head replacement (*A)

It displays the calculated date of cold head replacement, assuming 0-watt date of Average power or Heater Power as replacement date, calculated from approximation straight line which is set up from both end of sub graph display. This result is shown only in case of selecting Average Power or Heater Power in the Parameter list. 1st line shows the left end of the graph, 2nd line shows the right end of the graph, and 3rd line shows the calculated date assuming as this parameter becomes 0-watt.

Selected lines are copiable by "Ctrl + C" shortcut.

This list appears in opening any Mag log except with TN150 Revision none.

"#INFINITY" colored red appears as calculated result, in case the slope of approximation straight line was positive or 0. And, "#LESS" colored red appears as calculated result, in case of single point data. Further, this list blanks in case Average Power or Heater Power was not selected even if this list was selected in tab.

Calc	Raw	Point
Title	Date	Power
From	2017/11/26 02:00:31	0.79 [W]
To	2018/12/09 02:00:33	0.18 [W]
Result	2019/03/30 14:59:35	0.00 [W]

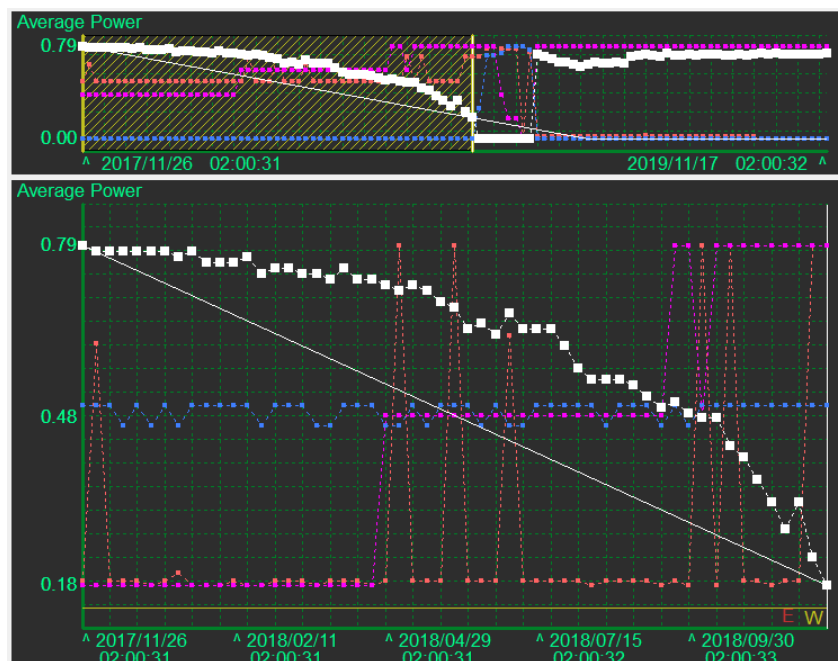
Calc	Raw	Point
Title	Date	Power
From	2019/03/17 02:00:31	0.63 [W]
To	2019/05/12 02:00:31	0.70 [W]
Result	#INFINITY	#####

Calc	Raw	Point
Title	Date	Power
From	2019/04/07 02:00:31	0.65 [W]
To		
Result	#LESS	#####

Forecast list of time for cold head replacement

(Upside: A case example of normal calculation, downside-left: a case example not of negative slope, downside-right: a case example calculated in single point data)

If the 0-watt date of Average Power or Heater Power is calculated normally, and the result is displayed in this list, then the approximation straight line appears on both graph area.



The calculated approximation straight line (thin line, take care that color depends on the opened Mag log.)

✓ Magnet field history list (*A)

It displays the history list of Magnet field. This list appears on whole graph area, in case checked the check-box at right-top of sub graph.

Selected lines are copiable by “Ctrl + C” shortcut.

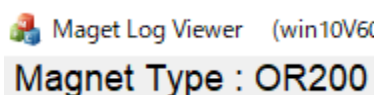
This list appears only in opening OR200 Mag log.

Magnet Type : OR200				Magnet Field ☒
No	Day	Time	Field	
053	2020/03/29	06:58:46	OFF	
054	2020/03/29	07:58:59	OFF	
055	2020/03/29	08:24:17	ON	
056	2020/03/29	08:59:12	OFF	
057	2020/03/29	09:59:24	OFF	
058	2020/03/29	10:59:36	OFF	
059	2020/03/29	11:59:46	OFF	
060	2020/03/29	12:59:58	OFF	
061	2020/03/29	14:00:09	OFF	
062	2020/03/29	15:00:18	OFF	
063	2020/03/29	16:00:28	OFF	
064	2020/03/29	17:00:39	OFF	
065	2020/03/29	18:00:49	OFF	
066	2020/03/29	19:01:03	OFF	
067	2020/03/29	20:01:13	OFF	
068	2020/03/29	21:01:25	OFF	
069	2020/03/29	22:01:35	OFF	
070	2020/03/29	23:01:46	OFF	
071	2020/03/30	00:01:58	OFF	
072	2020/03/30	01:02:09	OFF	
073	2020/03/30	02:00:18	ON	
074	2020/03/30	02:02:24	OFF	
075	2020/03/30	03:58:08	OFF	
076	2020/03/30	04:58:18	OFF	
077	2020/03/30	05:58:30	OFF	
078	2020/03/30	06:58:40	OFF	

Magnet field list (it exists check-box for change display between list and graph on right-top.)

✓ Magnet name display (*A)

It displays the magnet name corresponding with opening current Mag log.

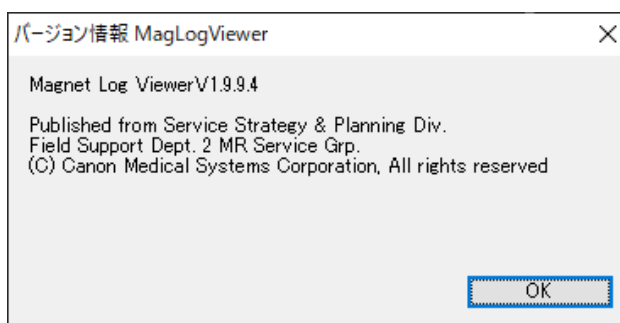


Magnet name display

✓ Version information

Software version and copyright notice of invoking Magnet Log Viewer is displayed in pop-up dialog.

This pop-up dialog can be shown selecting "Version information Magnet Log Viewer (&A)..." on displayed tooltip by clicking icon at left of title bar.



Version information

4.6. Close

Clicking "X" button on the right-top of window can close Magnet Log Viewer.

Typing "Alt + F4" is also available to close.

4.7. Uninstall

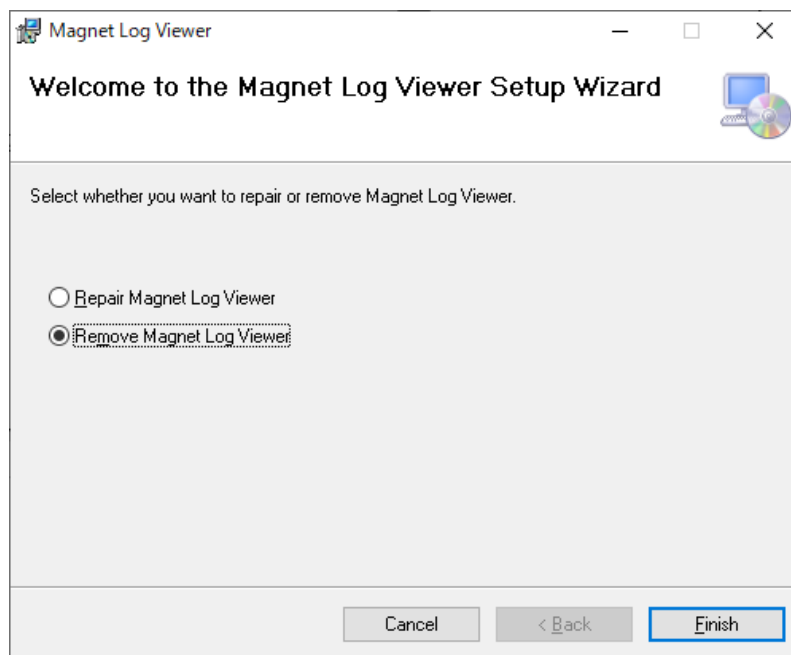
Magnet Log Viewer can be uninstalled two ways.

- ✓ MagLogViewer_JP.msi / MagLogViewer_EN.msi (below, MagLogViewer.msi)
- ✓ Windows application management

Below explain steps for each way.

✓ Uninstall from CoolCabiLogViewer.msi

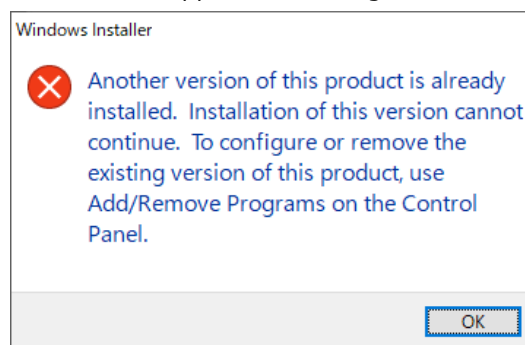
- (1) Execute MagLogViewer.msi on PC which Magnet Log Viewer has already been installed, after select "Remove Magnet Log Viewer"^(*A) on displayed dialog, and click [Finish]^(*A) on right-bottom of dialog.



Uninstallation entry screen of Magnet Log Viewer

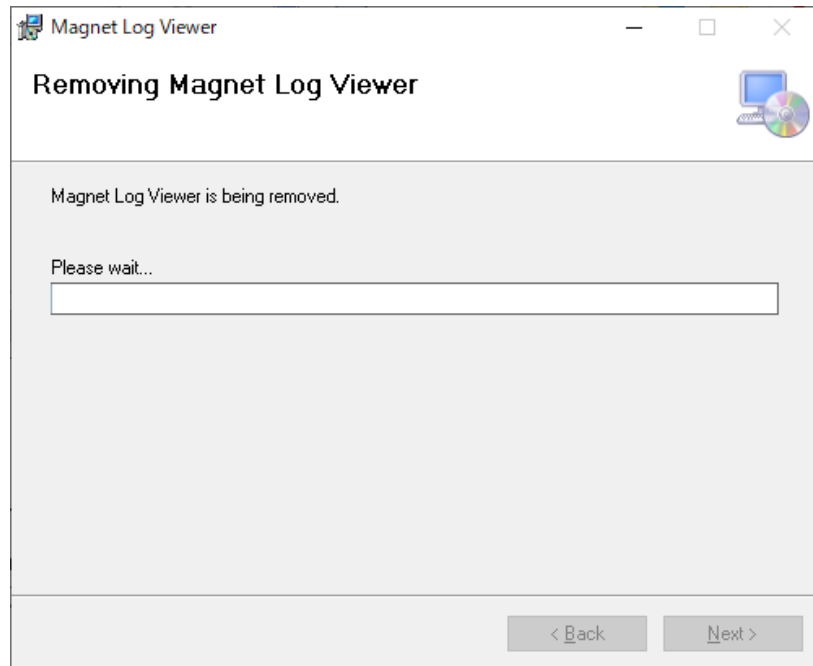
NOTE

Below dialog may be displayed, if the version between MagLogViewer.msi and installed Magnet Log Viewer is different. In this case, uninstall it following steps explained "Uninstall from Windows application management".



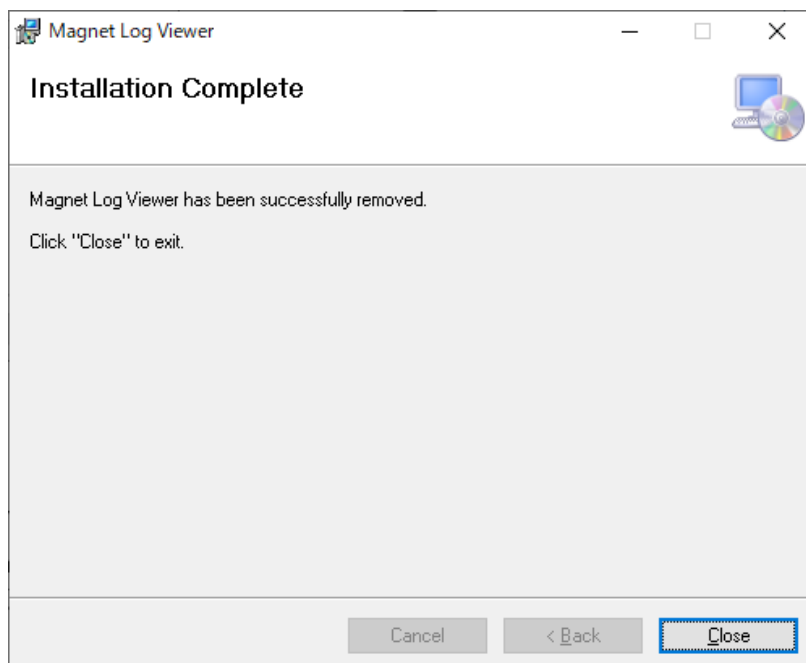
An example in case of unavailable uninstall from MagLogViewer.msi

- (2) After starting uninstallation, below screen is displayed, and progress bar under “Please wait...” message proceeds. Progress bar may be updated in several times, so please wait for a while.



Uninstallation in progress screen

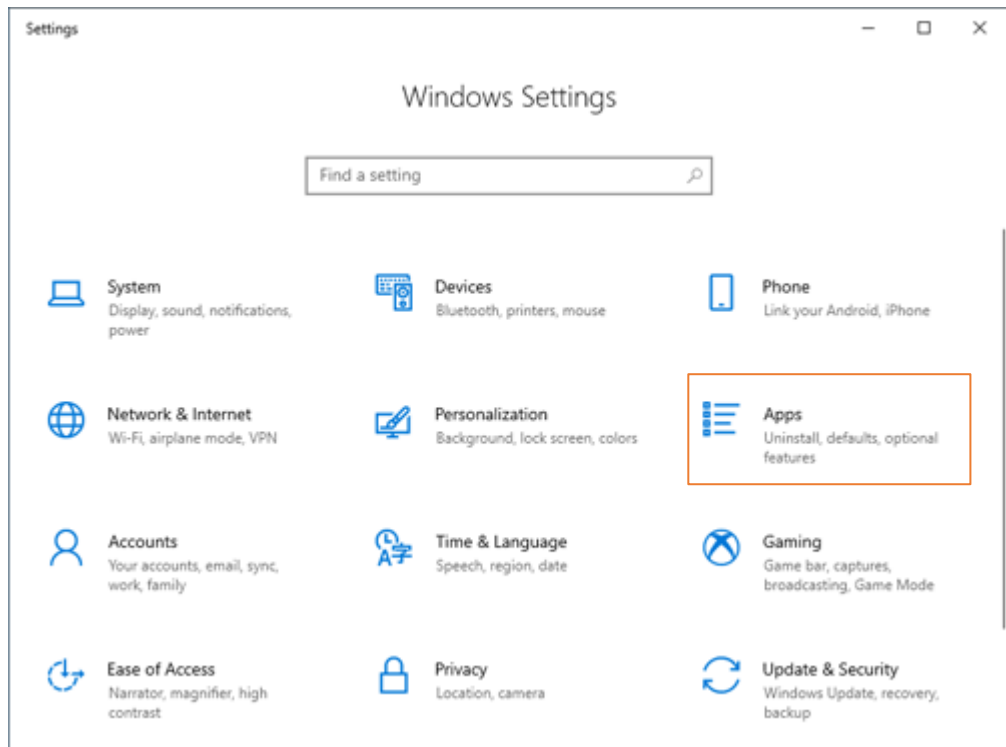
- (3) If below screen is displayed, uninstallation has been done successfully, so click [Close]^(*A) on right-below of dialog, and finish uninstallation.



Uninstallation success screen

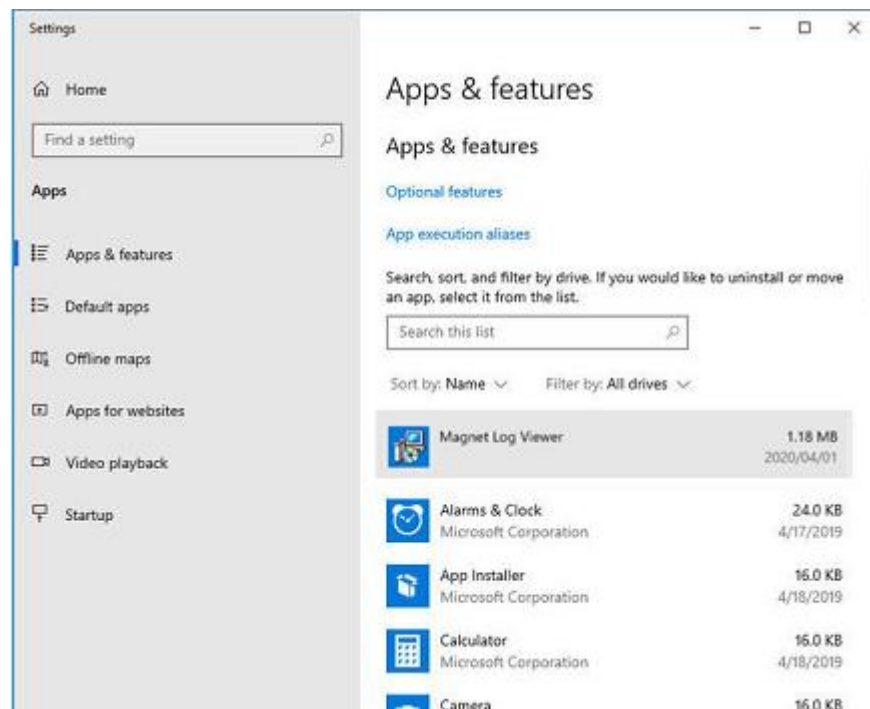
✓ **Uninstall from Windows application management**

(1) Open “Windows settings” dialog from windows start menu etc., select “Apps”.



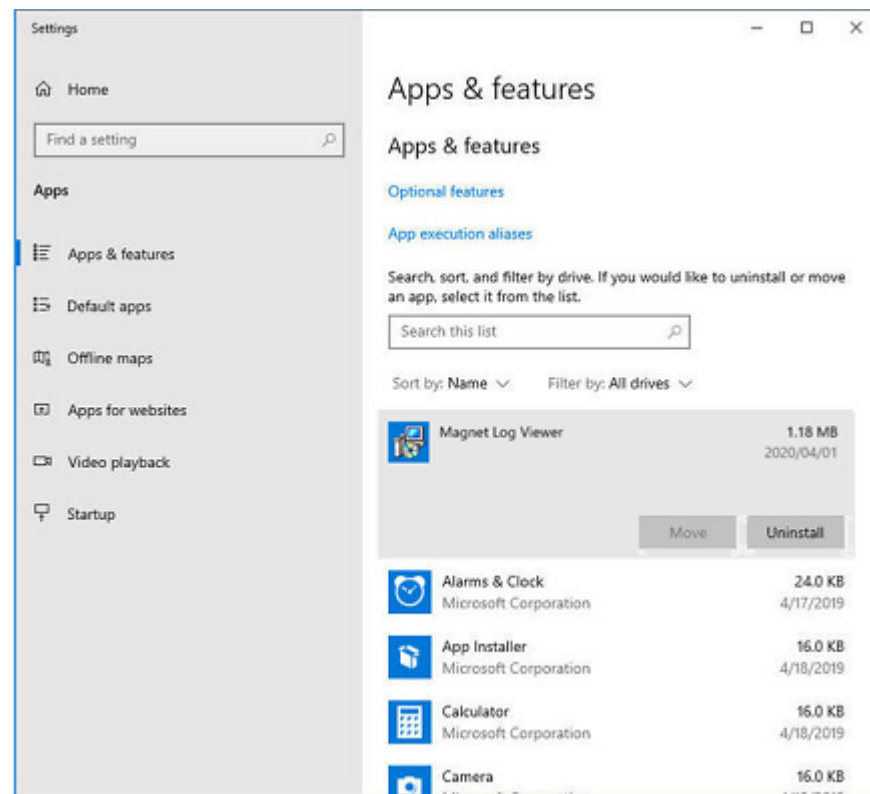
“Windows settings” screen

(2) Find and click “Magnet Log Viewer” from installed application list bottom of “Apps & features” screen.



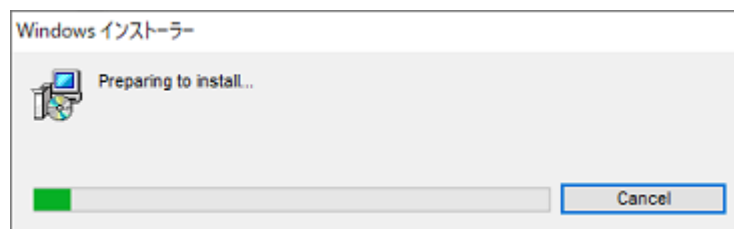
Magnet Log Viewer which is displayed in installed application list

- (3) Clicking “Uninstall” inside the selected frame, a pop-up dialog is displayed, so click “Uninstall” inside the pop-up dialog.



Screen selecting “Uninstall” on Magnet Log Viewer

- (4) Just after displayed pop-up dialog below, uninstall will be started, so please wait for a while. If it is completed, this pop-up dialog will disappear automatically.



Pop-up dialog displayed during execution uninstall

5. Notes

- ✓ Operation, display, and design of Magnet Log Viewer may change in the future without any announcement. However, an announcement will be made in cases of major important changes.
- ✓ If In the case of a defect report or a request for improvement for is required, please contact CMSC MR Service Group.